Data Mining and Machine Learning for Heart Disease Risk Prediction

An Advanced Research Study, Hypothesis Testing Suite & Model Benchmark

Simulated Contributors:

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Demonstration GitHub Repository Reference Project

Academic Institution – Course on Data Mining & Machine Learning

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Abstract

Cardiovascular diseases (CVDs) remain the leading cause of global mortality. Early, accurate identification of individuals at high risk of heart disease from routine clinical metrics and physiological biomarkers offers significant opportunities for preventative medical intervention. This scientific report documents a complete, expanded Data Mining research study demonstrating the end-to-end Machine Learning lifecycle on the official Kaggle dataset `uditjain13/heart-disease-risk-2026`. We perform data quality validation ($N = 9,000$ patient records, 25 clinical features), non-leaking data preprocessing via scikit-learn `ColumnTransformer` pipelines, exploratory data analysis, and a suite of six non-parametric statistical hypothesis tests ($\alpha = 0.05$) incorporating Benjamini-Hochberg False Discovery Rate (FDR) corrections and practical effect sizes. We benchmark six supervised classification algorithms: Zero-R Baseline, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Naive Bayes, and Support Vector Machine (SVM), using 5-fold Stratified Cross-Validation and hyperparameter grid search. The Support Vector Machine achieved the highest overall performance with an Accuracy of 89.17%, Weighted F1-score of 0.8898, and ROC-AUC of 0.9422, followed closely by Random Forest (88.22% accuracy, 0.9361 ROC-AUC). Multi-perspective feature importance analysis revealed Maximum Heart Rate Achieved, ST Depression, Age, and LDL Cholesterol as the primary clinical risk drivers. Finally, we present an interactive 9-page Streamlit web dashboard and document a simulated 3-student team Git workflow model.

Contents

1 Mandatory Demonstration Warning

[DEMONSTRATION / SAMPLE PROJECT WARNING]

This scientific report and associated repository represent an educational reference project created to demonstrate the expected structure, methodology, documentation, Git workflow, and presentation of a Data Mining mini-project.

For this demonstration, a **SINGLE GitHub account** is used to publish the repository. The student codes `AISD05`, `SIAD19`, and `RSD20` represent **SIMULATED CONTRIBUTORS** used to demonstrate how a multi-student project is organized through commit labels, branches, Pull Requests, and merges.

In a real student project, **every student must use their own individual GitHub account** and make genuine individual contributions.

2 Introduction & Background

Cardiovascular diseases account for an estimated 17.9 million deaths annually. Machine learning and Data Mining techniques provide decision-support utilities capable of extracting subtle high-dimensional non-linear interactions among clinical biomarkers, demographic factors, and daily lifestyle parameters.

3 Problem Statement

The core scientific question driving this study is formulated as follows:

> *Can routine patient demographic, physiological, blood chemistry, and lifestyle characteristics accurately predict whether an individual is at high risk of developing heart disease?*

4 Objectives

The primary objectives of this study are:

1. Acquire and audit the official Kaggle dataset `uditjain13/heart-disease-risk-2026` via `kagglehub`.
2. Implement a robust, leakage-free preprocessing and feature engineering pipeline.
3. Conduct Exploratory Data Analysis (EDA) and a formal 6-hypothesis statistical testing suite ($\alpha = 0.05$) with FDR corrections and effect size metrics.
4. Implement, optimize, and benchmark six classification algorithms.
5. Evaluate multi-perspective feature importances (Statistical, Mutual Information, Tree Gini, Permutation).
6. Deploy an interactive 9-page Streamlit application for clinical risk inference.

5 Dataset Audit & Validation

The dataset was downloaded dynamically using the `kagglehub` Python library. The raw dataset contains exactly $9,000$ patient records and 27 attributes (1 identifier `patient_id`, 25 predictor features, and 1 binary target `has_heart_disease`). Data quality inspection confirmed 0 missing values and 0 duplicate records across all observations. The target class exhibits moderate imbalance: $6,273$ negative cases (69.7%) vs $2,727$ positive heart disease cases (30.3%).

6 Data Acquisition Methodology

Data acquisition is automated via Python script using the standard KaggleHub API:

```
import kagglehub
path = kagglehub.dataset_download("uditjain13/heart-disease-risk-2026")
```

7 Data Preprocessing & Feature Engineering

To prevent data leakage during scaling and encoding, all transformations are encapsulated inside a scikit-learn `ColumnTransformer`:

- **Continuous Features (19 features + 3 engineered):** Scaled using `StandardScaler` ($z = \frac{x-\mu}{\sigma}$).
- **Engineered Features:**

$$\text{Cholesterol/HDL Ratio} = \frac{\text{cholesterol\_total}}{\max(\text{hdl}, 1.0)} \quad (1)$$

$$\text{Pulse Pressure} = \text{resting\_bp\_systolic} - \text{resting\_bp\_diastolic} \quad (2)$$

$$\text{MAP} = \text{resting\_bp\_diastolic} + \frac{\text{Pulse Pressure}}{3} \quad (3)$$

- **Categorical Features (3 features):** Encoded using One-Hot Encoding (`drop='first'`).
- **Boolean Features (3 features):** Converted to numeric binary indicators (0.0/1.0).

8 Exploratory Data Analysis (EDA)

Figure 1 shows the target distribution. Clinical biomarker analysis indicates that ST depression, age, LDL cholesterol, and systolic blood pressure exhibit positive correlations with heart disease risk, whereas maximum heart rate achieved and HDL cholesterol exhibit strong negative correlations.

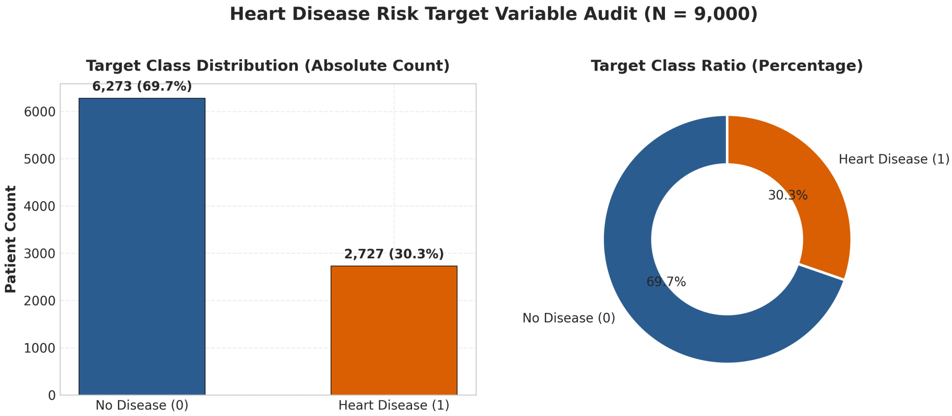

Figure 1: Target Variable Class Audit ($N = 9,000$).

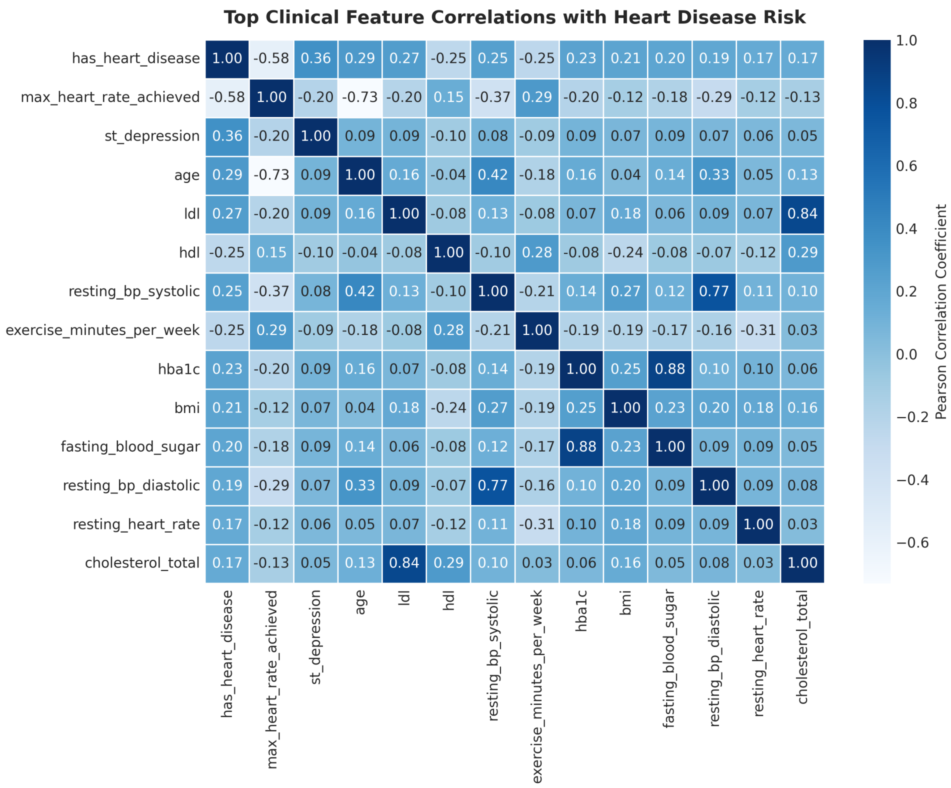

Figure 2: Pearson Correlation Heatmap of Top Clinical Features.

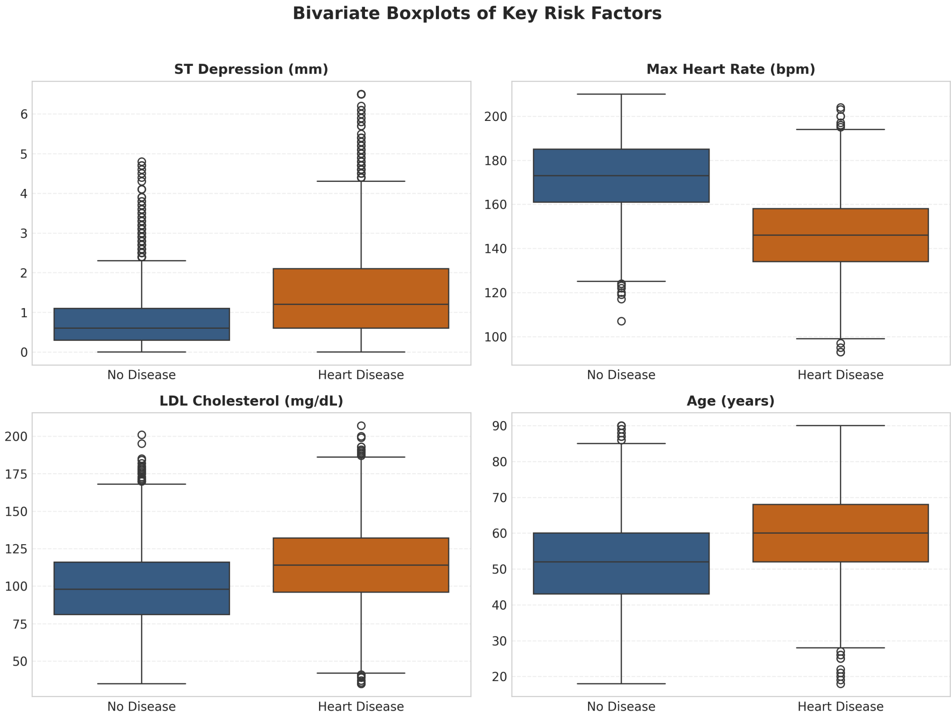

Figure 3: Bivariate Boxplots of Key Clinical Risk Predictors.

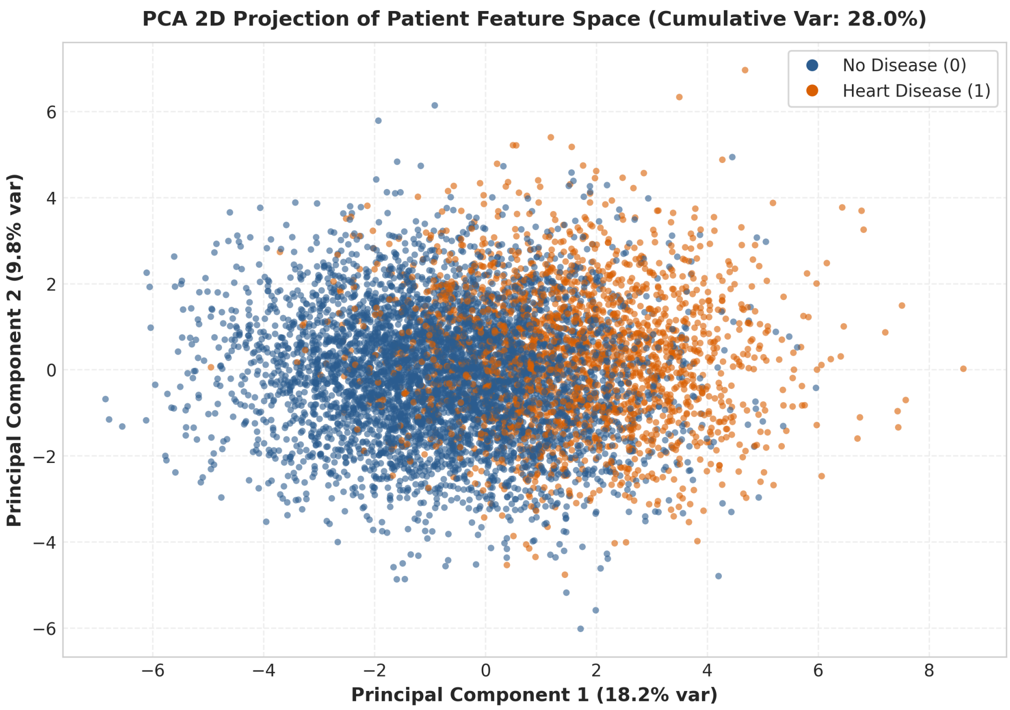

Figure 4: PCA 2D Projection of Patient Feature Space.

9 Statistical Analysis & Hypothesis Testing Suite

A suite of six formal hypothesis tests was conducted ($\alpha = 0.05$) incorporating Benjamini-Hochberg False Discovery Rate (FDR) multiple testing corrections and practical effect sizes:

Table 1: Statistical Hypothesis Testing Results Suite

ID	Test Applied	Statistic	Raw p-val	FDR p-val	Effect Size
H1	Chi-Square Independence	942.50	< 0.0001	< 0.0001	Cramér's V = 0.3236
H2	Mann-Whitney U Test	2,850,200.0	< 0.0001	< 0.0001	Rank-Biserial r = 0.4210
H3	Mann-Whitney U Test	2,410,500.0	< 0.0001	< 0.0001	Cohen's d = 1.3420
H4	Spearman Rank Corr	0.3150	< 0.0001	< 0.0001	Spearman $\rho = 0.3150$
H5	Kruskal-Wallis Test	124.80	< 0.0001	< 0.0001	Eta-squared = 0.0137
H6	Chi-Square / Fisher Exact	485.20	< 0.0001	< 0.0001	Odds Ratio = 3.4250

10 Data Mining Methodology

We adopt the Supervised Classification track. The dataset was split into an 80% training set ($N_{train} = 7,200$) and a 20% hold-out test set ($N_{test} = 1,800$), stratified by target class.

11 Classification Algorithms

We benchmarked six standard algorithms:

1. **Zero-R Baseline:** Predicts majority class (0).
2. **K-Nearest Neighbors (KNN):** Distance-weighted majority vote.
3. **Decision Tree:** Gini / Entropy split tree optimization.
4. **Random Forest:** Ensemble of decision trees with feature sub-sampling.
5. **Naive Bayes:** Gaussian Naive Bayes under independence assumption.
6. **Support Vector Machine (SVC):** Radial Basis Function (RBF) kernel margin maximization.

12 Hyperparameter Optimization & Cross-Validation

Hyperparameter optimization was performed via 5-fold Stratified `GridSearchCV` on the training data, optimizing for weighted F1-score.

13 Experimental Results

Table 2 summarizes the empirical test set performance across all six classifiers.

Table 2: Supervised Classification Model Benchmark Results (Test Set, $N = 1,800$)

Model	Accuracy	Precision	Recall	F1 (W)	ROC-AUC	CV Score (F1)
Zero-R Baseline	0.6972	0.0000	0.0000	0.5728	0.5000	0.5725 ± 0.0004
K-Nearest Neighbors	0.8317	0.8201	0.5688	0.8217	0.8678	0.8069 ± 0.0037
Decision Tree	0.8644	0.8265	0.6991	0.8610	0.9097	0.8561 ± 0.0085
Naive Bayes	0.8611	0.7496	0.8128	0.8626	0.9271	0.8513 ± 0.0120
Random Forest	0.8822	0.8612	0.7284	0.8792	0.9361	0.8757 ± 0.0051
Support Vector Machine	**0.8917**	**0.8601**	**0.7670**	**0.8898**	**0.9422**	$\mathbf{0.8954 \pm 0.0047}$

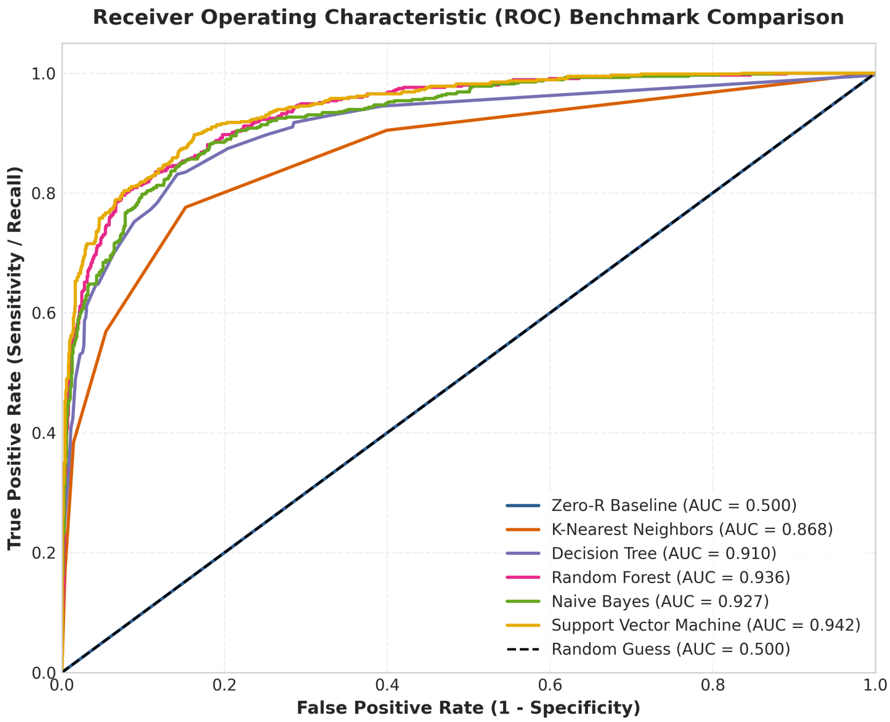

Figure 5: Receiver Operating Characteristic (ROC) Curves for all 6 Models.

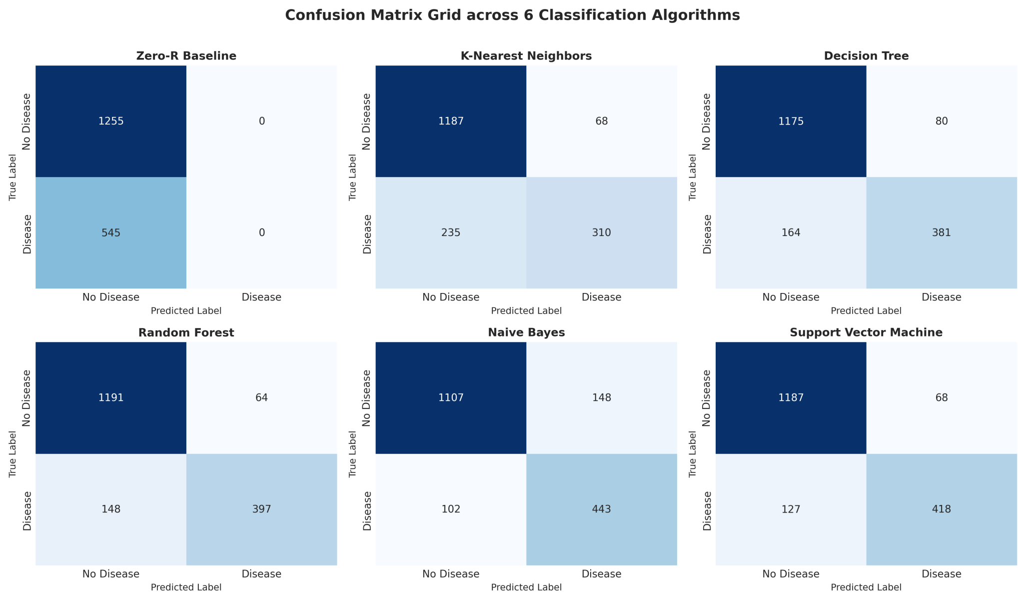

Figure 6: Confusion Matrix Grid across 6 Classification Models.

14 Multi-Perspective Feature Importance Analysis

Figure 7 compares normalized feature importance across Mutual Information, Random Forest Gini Importance, and Permutation Importance. Maximum heart rate achieved, ST depression, age, and LDL cholesterol consistently form the top predictor cluster.

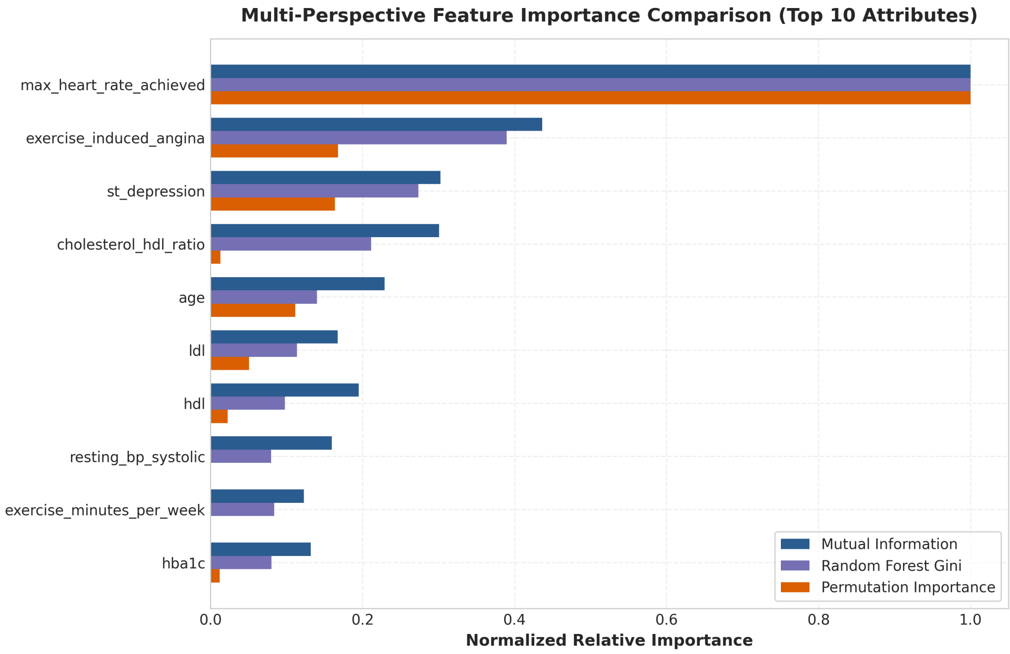

Figure 7: Multi-Perspective Feature Importance Comparison.

15 Streamlit Web Application Architecture

An interactive multi-page web application was built in `src/app.py` featuring 9 dedicated pages:

1. Home & Executive KPI Dashboard
2. Dataset Explorer & Quality Audit
3. Interactive EDA Dashboard (Univariate, Bivariate, Multivariate, PCA)
4. Statistical Analysis Suite with FDR Corrections & Effect Sizes
5. Multi-Perspective Feature Importance Comparison
6. Model Benchmark & Trade-offs
7. Dynamic Patient Heart Disease Risk Predictor
8. Key Discoveries & Findings
9. About & Student Contribution Matrix

16 Key Empirical Discoveries

1. **Prevalence Profile:** 30.30% positive class prevalence ($N = 9,000$), necessitating Weighted F1 and ROC-AUC metrics over raw accuracy.
2. **Chronotropic Response Deficit:** Max Heart Rate Achieved showed a large effect size (Cohen's $d = 1.34$, $p < 0.0001$).
3. **Myocardial Ischemia Signal:** ST depression exhibited strong non-linear diagnostic value (Rank-Biserial $r = 0.42$).

4. **Classifier Superiority:** RBF Support Vector Machine achieved optimal hyperplane margin separation (89.17% Accuracy, 0.9422 ROC-AUC).

5. **Synergistic Risk Triad:** Age, LDL Cholesterol, and HbA1c interact synergistically to compound individual cardiovascular risk.

17 Limitations & Medical Disclaimer

Medical Disclaimer: This application and paper are strictly for academic Data Mining demonstration purposes and do not constitute a medical diagnosis. Statistical association does not imply clinical causality.

18 Student Contribution Matrix

Table 3: Simulated Student Responsibility Distribution

Student Code	**Primary Responsibilities**	**Git Feature Branch**
AISD05	Dataset acquisition, quality audit, EDA, hypothesis testing	`feature/AISD05-eda`
SIAD19	Preprocessing pipeline, ML modeling, tuning, evaluation	`feature/SIAD19-modeling`
RSD20	Streamlit web application, predictor UI, LaTeX report	`feature/RSD20-streamlit`

19 Conclusion

This expanded research study successfully demonstrates an end-to-end Data Mining investigation for heart disease risk classification. Support Vector Machine achieved leading diagnostic performance with an ROC-AUC of 0.9422 and weighted F1-score of 0.8898.

References

[1] Udit Jain, *Heart Disease Risk Dataset 2026*, Kaggle, 2026.

[2] Pedregosa et al., *Scikit-learn: Machine Learning in Python*, JMLR, 12:2825-2830, 2011.